

Sales & Market Share Dashboard

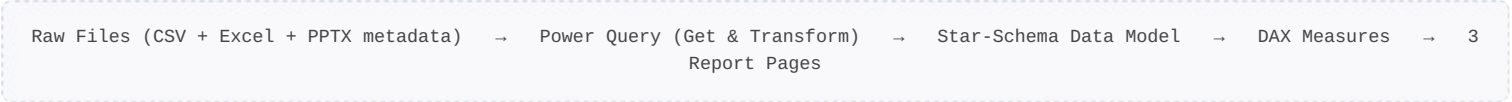
Power BI Case Study — Project Documentation

End-to-end build: [Data Integration](#) → [Modeling](#) → [DAX](#) → [Visualization](#)

A full walkthrough of how this Power BI report was built — from raw source files through Power Query transformation, star-schema data modeling, DAX measures, and three finished report pages — including the reasoning behind each design and charting decision.

1. Project Overview

This project is a Power BI case study simulating a sales and market share analysis for a manufacturing business. The brief assumes the report's owner is one of several competing manufacturers (**Atlas**), and the deliverable analyzes both Atlas's internal performance and its position relative to its 13 competitors, across five countries and multiple years of transaction data.



Source files provided

File	Purpose
Assignment_Case_Study.docx	Project brief and requirements
Metadata.pptx	Data dictionary describing every source column
Proposed_Design_and_theme.PNG	UI/theme reference the final report was styled against
Sales.csv	Domestic (USA) sales transactions
Canada.csv, Germany.csv, Japan.csv, Mexico.csv, Nigeria.csv	International sales transactions, one file per country
bi_dimensions.xlsx	Three reference sheets: Product Details, Manufacturer, Geography

2. Data Integration & Transformation (Power Query)

Every source was brought in through Power Query, cleaned, and shaped into five final tables. Below is what each source needed and why.

2.1 Fact_Sales (the central fact table)

Built by combining two separate imports:

- Sales.csv** (domestic) — imported directly. No `Country` column existed in this file, so a custom column was added: `Country = "USA"`, matching the exact spelling used in the Geography dimension.
- International Sales folder** (Canada, Germany, Japan, Mexico, Nigeria) — imported using Power Query's **Get Data** → **Folder** → **Combine & Transform Files**, which auto-generates a reusable transform function and appends all five country files (each already contained a `Country` field).

The two were combined with **Append Queries** into one fact table of roughly 1.6 million+ rows.

Data quality fix: `Zip` was initially auto-typed as a Whole Number. This silently strips leading zeros from US zip codes (e.g. `02138` → `2138`) and breaks entirely for alphanumeric zips in Canada/Japan. Fixed by explicitly setting `Zip` to **Text** at every relevant step, and by parsing `Date` with an explicit **en-US locale** to avoid day/month misparsing.

A composite key, `ZipCountryKey = [Zip] & "-" & [Country]`, was added to support joining to Geography (see §3).

2.2 Dim_Product (from bi_dimensions.xlsx → "Product Details")

- Removed a stray title row above the real header, then promoted headers.
- Category** was only populated on the first row of each group (a merged-cell pattern from Excel) — fixed with **Fill Down**.
- Price** was stored as text like `"USD 412.13"` — split by space into `Currency` and a numeric `Price` column.

2.3 Dim_Manufacturer (from "Manufacturer" sheet)

This sheet was stored as a **transposed cross-tab**: 3 rows (ManufacturerID / Manufacturer / Logo) by 14 columns (one per manufacturer) — the opposite orientation of a normal table.

- Used **Transform** → **Transpose** to flip it into 15 rows × 3 columns.
- Promoted the new first row as headers, producing a clean `ManufacturerID | Manufacturer | Logo` table of 14 manufacturers.

2.4 Dim_Geography (from "Geography" sheet — 176,931 rows)

- Removed 3 junk rows above the real header (`Source:` , `Last Updated:` , a blank row).
- Region** had scattered nulls (mostly Mexico rows) → replaced with `"Unassigned"` .
- Country** was blank on exactly 2 rows → investigated the rows (State = "Ondo" / "Oyo", both Nigerian states) and filled with `"Nigeria"` rather than dropping the rows.
- City** was blank on ~18,500 rows (~10%, mostly Mexico) → fallback logic: use `District` when City is blank, else `"Unassigned"` .
- Rows with a blank `Zip` were filtered out (a Zip-less geography row can never match a sale).
- Built the same `ZipCountryKey` as Fact_Sales, then applied **Remove Duplicates** on that key to guarantee uniqueness — required because Power BI relationships need a unique key on the "one" side.

Why not just join on Zip alone? Zip codes are not globally unique — a numeric zip in the US can collide with a postal code elsewhere. The composite `Zip +`

`Country` key avoids incorrect cross-country matches.

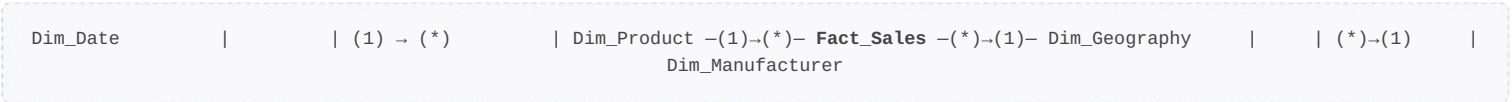
2.5 Dim_Date (built from scratch — no date table was supplied)

Generated with a blank Power Query referencing `List.Min / List.Max` of `Fact_Sales[Date]` , so the table always spans exactly the data's real date range. Added calculated columns: `Year` , `Quarter` , `MonthNumber` , `MonthName` , `YearMonth` , and later `YearMonthNumber` (`Year × 100 + MonthNumber`) as a numeric sort key so month labels display in chronological order rather than alphabetical order in charts.

Marked as the model's official **Date Table** (required for the time-intelligence DAX functions used later), and `MonthName` is sorted by `MonthNumber` .

3. Data Model & Relationships

The model follows a **star schema**, with one small **snowflake** branch for Manufacturer (a legitimate, common pattern — Manufacturer is an attribute one step removed from Product, not a direct fact-table attribute).



From (one side)	To (many side)	Key
Dim_Product[ProductID]	Fact_Sales[ProductID]	ProductID
Dim_Manufacturer[ManufacturerID]	Dim_Product[ManufacturerID]	ManufacturerID (snowflake)
Dim_Date[Date]	Fact_Sales[Date]	Date
Dim_Geography[ZipCountryKey]	Fact_Sales[ZipCountryKey]	Zip + Country composite

All relationships are single-direction, one-to-many, filtering from dimension to fact — the standard, most performant configuration and the one the brief's "tidy and efficient data model" criterion is checking for.

Why Manufacturer wasn't merged directly into Product

Keeping Manufacturer as its own dimension lets it be used directly as a slicer/axis field without going through Product, and avoids repeating manufacturer text on every one of the 2,412 product rows. The alternative (flattening Manufacturer into Product) would also be valid — this was a design choice, not a requirement.

Model hygiene

- Join-only columns (`ProductID` , `ManufacturerID` , `Zip` , `ZipCountryKey`) are hidden from report view.
- All DAX measures live in a dedicated, empty `_Measures` table rather than scattered across data tables.

4. DAX Measures

4.1 Core measures

Measure	DAX	Purpose
Total Revenue	<code>SUM(Fact_Sales[Revenue])</code>	Base revenue aggregate used throughout the report
Total Units	<code>SUM(Fact_Sales[Units])</code>	Volume metric
Total Orders	<code>COUNTROWS(Fact_Sales)</code>	Transaction count
Avg Selling Price	<code>DIVIDE([Total Units], [Total Revenue])</code>	Blended price point

4.2 Market share & ranking

Measure	Purpose
Total Market Revenue	Revenue with the Manufacturer filter stripped (<code>ALL(Dim_Manufacturer)</code>), while respecting any active Year/Region/Category slicer — the denominator for market share
Market Share %	<code>DIVIDE([Total Revenue], [Total Market Revenue])</code> — a given manufacturer's slice of the currently-filtered market
Manufacturer Rank	<code>RANKX</code> of manufacturers by revenue within current filter context
Top 5 Manufacturer Flag	Flags the top 5 manufacturers <i>within whatever filter context a visual is in</i> — used for conditional formatting
Top 5 Manufacturer Flag (All-Time)	A context-safe version that strips <code>ALL(Dim_Date)</code> before ranking, so the same 5 manufacturers stay highlighted regardless of which Year is on a chart's axis — needed after the plain version gave inconsistent results on a Year-by-Manufacturer trend chart
Atlas Rank	Atlas's specific rank among all manufacturers, as a standalone KPI

4.3 Top N filtering (parameter-driven)

Backed by a **What-If Parameter** (`Top N Value` , numeric range 1–20, default 5):

Measure	Purpose
Top N Product Rank	<code>RANKX(ALL(Dim_Product), [Total Revenue], , DESC, DENSE)</code>
Top N Products Revenue	Revenue shown only when a product's rank ≤ the current slicer value
Top N Region Rank / Top N Regions Revenue	Same pattern, applied to Region

Moving the `Top N Value` slicer dynamically resizes the Top N charts — the core of Requirement II in the brief.

4.4 Time intelligence & competitor analysis

Measure	DAX / Logic	Purpose
Prior Year Revenue	CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[Date]))	Same-period-last-year comparison base
YoY Growth %	DIVIDE([Total Revenue]-[Prior Year Revenue], [Prior Year Revenue])	Year-over-year growth rate, conditionally colored green/red
Revenue Trend Status	IF/branching on YoY Growth %	"Growth" / "Decline" / "Flat" label
Company Revenue	Total Revenue filtered to Dim_Manufacturer[Manufacturer] = "Atlas"	Atlas-only revenue
Competitor Revenue	[Total Revenue] - [Company Revenue]	Everyone else's revenue
Company Market Share %	DIVIDE([Company Revenue], [Total Revenue])	Atlas's overall share of the market

4.5 Field Parameter

A Breakdown By Field Parameter was created (Manufacturer / Region / Category / Segment) to allow a single visual's axis to be swapped dynamically via a slicer, rather than building a separate chart per dimension.

5. Report Pages & Visualization Choices

Three pages, one per requirement in the brief, all sharing the same custom color theme (see §6).

Page 1 — Market Share Analysis Requirement I

Visual	Chart type	Why this type
KPI row (Total Revenue, Company Revenue, Company Market Share %, Atlas Rank)	Multi-value Card	Glanceable headline numbers before any interaction — the standard "first thing you read" pattern on a dashboard
Manufacturer table	Table, sorted descending by revenue	Precise numeric ranking with exact figures — a bar chart alone can't show exact percentages as cleanly as a sortable table
Total Revenue by Manufacturer	Clustered bar chart, with a Manufacturer → Region → Category drill hierarchy	Horizontal bars make 14 manufacturer-name labels easy to read (vs. cramped vertical columns); the drill hierarchy lets a viewer click into any manufacturer's geographic/category composition without leaving the chart
Top 5 highlighting	Conditional data-color formatting driven by the Top 5 flag measure	Keeps all 14 manufacturers visible for full competitive context, while still visually calling out the leaders — chosen over hard-filtering down to 5, which would hide the rest of the field
Year / Region / Category slicers	Tile (Year) + dropdown (Region, Category)	Satisfies the brief's explicit "filter by year, region, and product category" requirement

Page 2 — Top N Filtering Requirement II

Visual	Chart type	Why this type
Top N Value	What-If Parameter slider	Lets the viewer choose N interactively rather than being locked to a hardcoded "top 5"
Top Products by Revenue	Column chart	Vertical columns read naturally left-to-right for a small, ranked product list
Top Regions by Revenue	Column chart	Same reasoning, applied to Region

Page 3 — Performance Comparison Requirement III

Visual	Chart type	Why this type
Total Revenue by Month	Area chart	Matches the provided design reference exactly; the filled area emphasizes trend shape over time better than a bare line
YoY Growth %	Card, conditional color (green/red)	A single number with color is the fastest way to communicate "up or down" at a glance
Atlas vs. Competitors — Revenue Share	100% stacked bar chart	Frames the comparison as "share of one whole," which is the clearest way to show a small company against a large competitor pool
Revenue Trend by Top 5 Manufacturers	Line chart, one line per manufacturer	Line charts are the standard choice for comparing multiple series' trajectories over time; filtered to the (all-time) top 5 to stay readable
Explore Revenue Drivers	Decomposition Tree	Lets a viewer interactively explore Total Revenue across Manufacturer, Region, and Category in any order, in a single visual — a more advanced, exploratory alternative to building separate static charts for every dimension combination

6. Design & Theme

A custom Power BI theme was built to match the provided reference image, rather than using a default palette:

#5F7DF5 Primary blue	#683CEC Purple accent	#EF546F Red / decline
#F5F9FA Page background	#F9F9F9 Card background	#2A2B29 Text

- All visuals sit inside rounded-corner white cards with a light border, matching the reference's card-based layout.
- Year slicer styled as horizontal "tile" pills to mirror the reference's fiscal-year tab row.
- KPI cards use a pastel icon chip (accent-tinted square) beside each number.

7. Data Quality Findings

Several real gaps and inconsistencies were found in the source data during the build. Rather than hiding them, they were investigated and handled explicitly:

- **Missing Country on domestic sales** — Sales.csv had no Country field at all; resolved by hardcoding "USA" , matched exactly to Geography's spelling convention.
- **Transposed Manufacturer sheet** — required a Transpose step rather than a standard import.
- **~10% of Geography rows missing City** (mostly Mexico) — filled from District, with an "Unassigned" fallback.

- **2 Geography rows missing Country** — inferred as Nigeria from their listed state names and filled rather than dropped.
- **Zip typed as a number** — a bug that silently lost leading US zeros and broke alphanumeric Canada/Japan codes; corrected to Text throughout.
- **Zip codes are not globally unique** — required a composite Zip+Country key for a valid one-to-many relationship.
- **Some sales rows likely don't match any Geography row** — surfaced as a "(Blank)" region bucket in early testing; flagged for verification against the Zip-typing fix above.

8. How This Maps to the Brief's Evaluation Criteria

Criterion	Where it's addressed
Data modelling approach	Star schema with one justified snowflake branch, all-1-to-many relationships, hidden join keys, marked Date table (\$3)
DAX correctness & performance	Dedicated measures table, context-safe ranking measures, All-Time flag fix for multi-dimension charts (\$4)
Visual clarity & usability	Consistent custom theme, card-based layout, sortable/highlighted views, dynamic parameter-driven filtering (\$5, \$6)
Analytical insights	Market share and rank framing throughout; explicit data-quality findings documented rather than hidden (\$7)
Overall solution design	Three pages directly mapped to the brief's three required analyses, consistent theme and interaction patterns across all of them